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1 Introduction

Unix provides a powerful set of command-line utilities that allow users to manage files, directories, and system resources efficiently. To understand these utilities is essential for directories, and system resources efficiently. These utilities is essential for effectively in UNIX-based Operating Systems such as Linux. This logbook focuses on practicing commonly used Unix command, such as cat, grep, alias, and history which are widely used for file handing, text processing, and command line productivity.

2 Aim

The aim of this logbook is to gain practical experience in using UNIX utilities for directory management, file creation, text processing, command aliasing, and command history manipulation.

3 Objectives

- To create and navigate directory structures using UNIX commands.
- To manage and create files using the cat command.
- To understand and apply different options of the grep utility.
- To define, use, remove, and preserve command aliases.
- To explore command history and re-execution features in UNIX.

4 Steps Configurations

4.1 Create the required Directory Structure

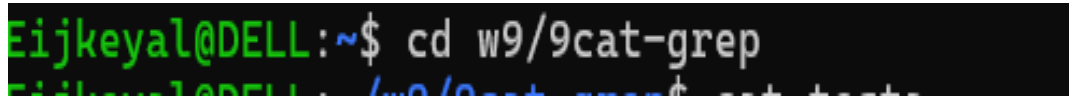
The required directory structure was created using mkdir command with the -p options to ensure parent directories were created automatically.

```
Eijkeyal@DELL:~$ script c11
Script started, output log file is 'c11'.
Eijkeyal@DELL:~$ cd $HOME
Eijkeyal@DELL:~$ mkdir -p ~/w9/9cat-grep
Eijkeyal@DELL:~$ ls
c11  File1  Folder  Obj1   test2  w7      w7w7-2  Workshop8
C11  File2  Laptop  test1  test3  w7w7-1  w9
Eijkeyal@DELL:~$ cd ~
```

Figure 1 Create Directory

4.2 Navigate Using a Relative Path

The `cd` command was used with a relative pathname to change into the `9cat-grep` directory, demonstrating efficient directories navigation.

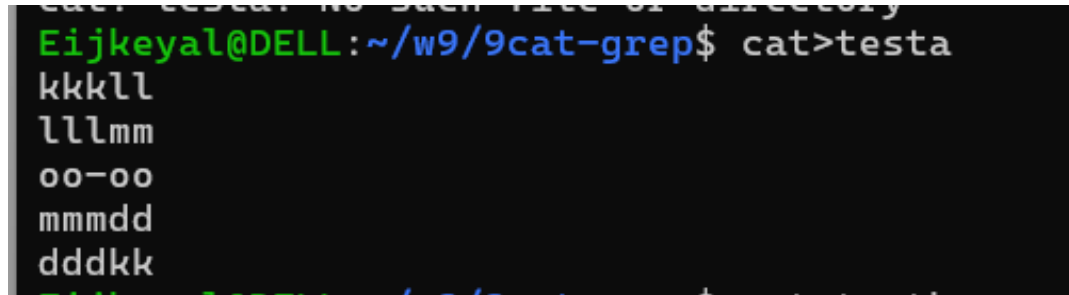


```
Eijkeyal@DELL:~$ cd w9/9cat-grep
```

Figure 2 Change to Relative Path

4.3 Create Files Using Cat

Two text files, `testa` and `testb`, were created using the `cat` utility. This demonstrated how standard input can be redirected to create files and store textual data.

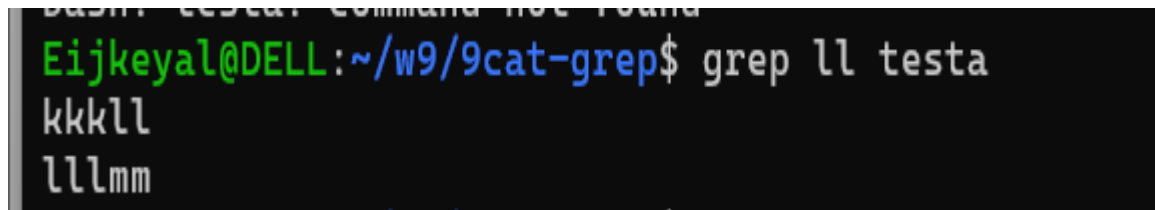


```
Eijkeyal@DELL:~/w9/9cat-grep$ cat>testa
kkkll
lllmm
oo-oo
mmmdd
dddkk
```

Figure 3 Creating Files Using Cat Command

4.4 Use of grep-utility

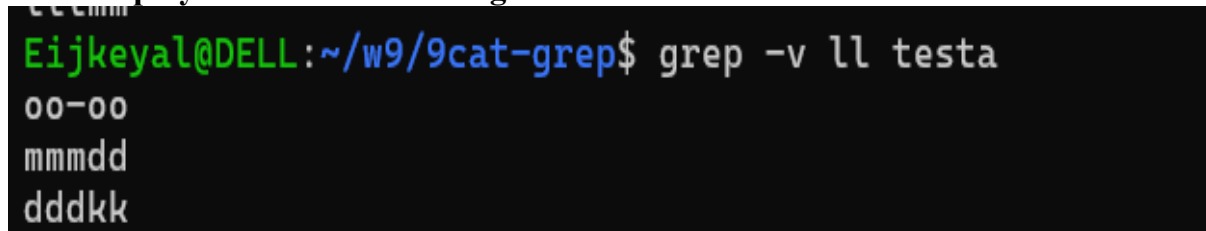
Displays lines containing `ll`.



```
Eijkeyal@DELL:~/w9/9cat-grep$ grep ll testa
kkkll
lllmm
```

Figure 4 Displaying common lines ll

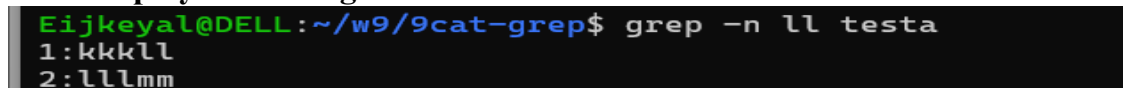
4.5 Displays lines not containing ll.



```
Eijkeyal@DELL:~/w9/9cat-grep$ grep -v ll testa
oo-oo
mmmdd
dddkk
```

Figure 5 Displaying which don't contain ll.

4.6 Displays matching lines with line numbers.



```
Eijkeyal@DELL:~/w9/9cat-grep$ grep -n ll testa
1: kkkll
2: lllmm
```

Figure 6 Displaying common lines with numbers.

4.7 Display File names

```
Eijkeyal@DELL:~/w9/9cat-grep$ grep -l ll *  
testa
```

Figure 7 Display file names

4.8 Case-insensitive search (ll, LL, Ll).

Same Result as above because of -i.

```
Eijkeyal@DELL:~/w9/9cat-grep$ grep -i ll *  
testa:kkkll  
testa:lllmm  
testb:LLLL
```

Figure 8 Display case insensitive.

4.9 Displays the count of matching lines per file.

```
Eijkeyal@DELL:~/w9/9cat-grep$ grep -c ll *  
testa:2  
testb:0
```

Figure 9 Display the count of meaning lines.

4.10 Displays Lines starting with K.

```
Eijkeyal@DELL:~/w9/9cat-grep$ grep '^K' testa testb
```

Figure 10 Display Lines start with k

4.11 Displays all lines with line numbers.

```
Eijkeyal@DELL:~/w9/9cat-grep$ grep -n '^' testa  
1:kkkll  
2:lllmm  
3:oo-oo  
4:mmmdd  
5:dddkk
```

Figure 11 Display all lines with line numbers

4.12 Define alias lsal.

```
Eijkeyal@DELL:~/w9/9cat-grep$ alias lsal='ls -al'  
Eijkeyal@DELL:~/w9/9cat-grep$ alias lsal  
alias lsal='ls -al'
```

Figure 12 Define alias lsal

Show stored alias and alias works only in the current session.

```
Eijkeyal@DELL:~/w9/9cat-grep$ cd ~
Eijkeyal@DELL:~$ lsal
total 112
drwx----- 11 Eijkeyal Eijkeyal 4096 Dec 19 06:07 .
drwxr-xr-x  3 root      root      4096 Dec  4 07:50 ..
-rw-----  1 Eijkeyal Eijkeyal 2877 Dec 18 12:46 .bash_history
-rw-r--r--  1 Eijkeyal Eijkeyal 220  Dec  4 07:50 .bash_logout
-rw-r--r--  1 Eijkeyal Eijkeyal 3574 Dec 18 08:04 .bashrc
-rw-r--r--  1 Eijkeyal Eijkeyal 3527 Dec 18 07:57 '.bashrc alias lsl='\''la -l'\''
-rw-r--r--  1 Eijkeyal Eijkeyal 3622 Dec 18 12:46 .bashrc.save
-rw-r--r--  1 Eijkeyal Eijkeyal 4096 Dec 19 06:20 c11
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 11 08:14 C11
-rw-r--r--  1 Eijkeyal Eijkeyal  0 Dec  4 08:04 File1
-rw-r--r--  1 Eijkeyal Eijkeyal  0 Dec  4 08:04 File2
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec  4 08:12 Folder
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec 11 07:33 Laptop
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 11 07:56 .local
drwxr-xr-x  2 Eijkeyal Eijkeyal 4096 Dec  4 08:03 Obj1
-rw-r--r--  1 Eijkeyal Eijkeyal 807  Dec  4 07:50 .profile
-rw-r--r--  1 Eijkeyal Eijkeyal  0 Dec  4 08:14 .sudo_as_admin_successful
-rw-r--r--  1 Eijkeyal Eijkeyal 24  Dec  5 06:00 test1
-rw-r--r--  1 Eijkeyal Eijkeyal 62  Dec  5 06:02 test2
-rw-r--r--  1 Eijkeyal Eijkeyal 86  Dec  5 06:04 test3
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 12 07:12 w7
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec 12 06:30 w7w7-1
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec 12 06:30 w7w7-2
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 19 06:07 w9
-rw-r--r--  1 Eijkeyal Eijkeyal 24501 Dec 12 18:21 Workshop8
```

Figure 13 Works only in the current session

4.13 Remove the alias.

```
Eijkeyal@DELL:~$ unalias lsal
```

Figure 14 Remove alias

4.14 Verify removal.

Alias no longer exists.

```
Eijkeyal@DELL:~$ alias lsal
bash: alias: lsal: not found
```

Figure 15 Alias no longer exist

4.15 Preserve alias for next session.

```
Eijkeyal@DELL:~$ echo "alias lsal='ls -al'" >> ~/.bashrc
Eijkeyal@DELL:~$ source ~/.bashrc
```

Figure 16 Store alias for next session

4.16 Alias persists after logout/login.

```
Eijkeyal@DELL:~$ alias lsal
alias lsal='ls -al'
```

Figure 17 Alias persists after logout/login

4.17 Define nwho alias.

Counts number of system user accounts.

```
Eijkeyal@DELL:~$ alias nwho='getent passwd | wc -l'
```

Figure 18 Counts number of system account

4.18 Execute nwho

Output may differ from each other users due to different system configurations.

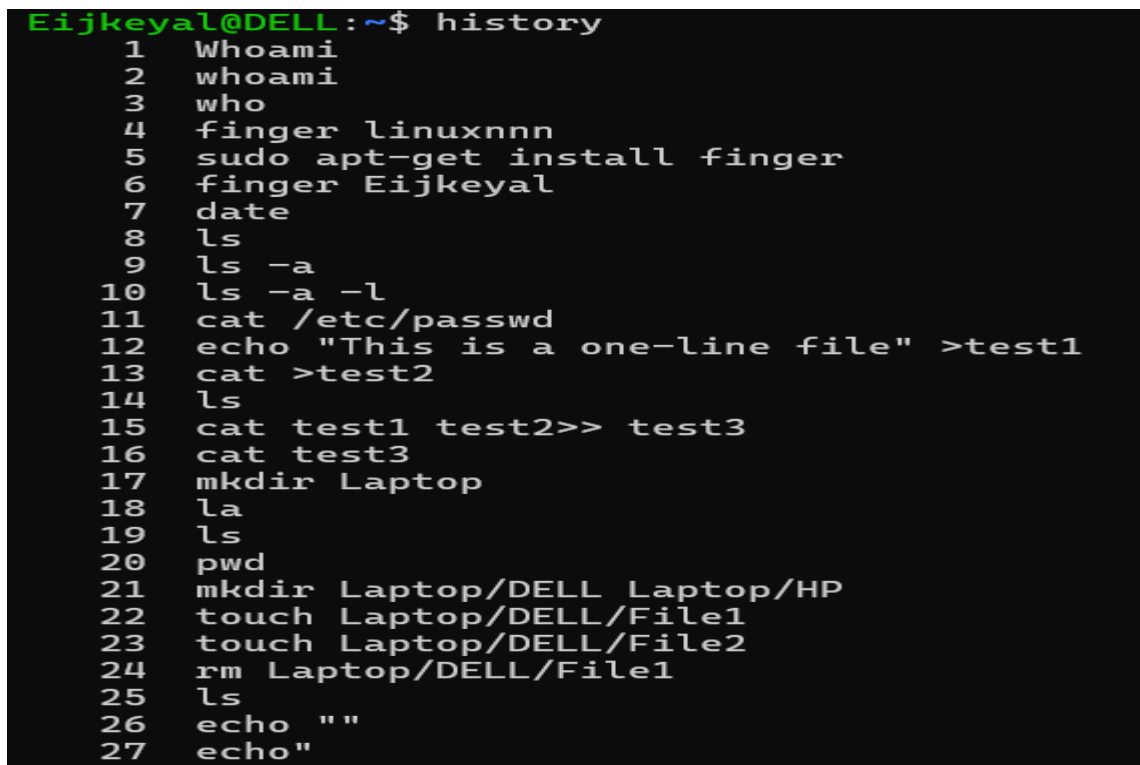


```
Eijkeyal@DELL:~$ nwho
22
```

Figure 19 Executes the nwho

4.19 List command history.

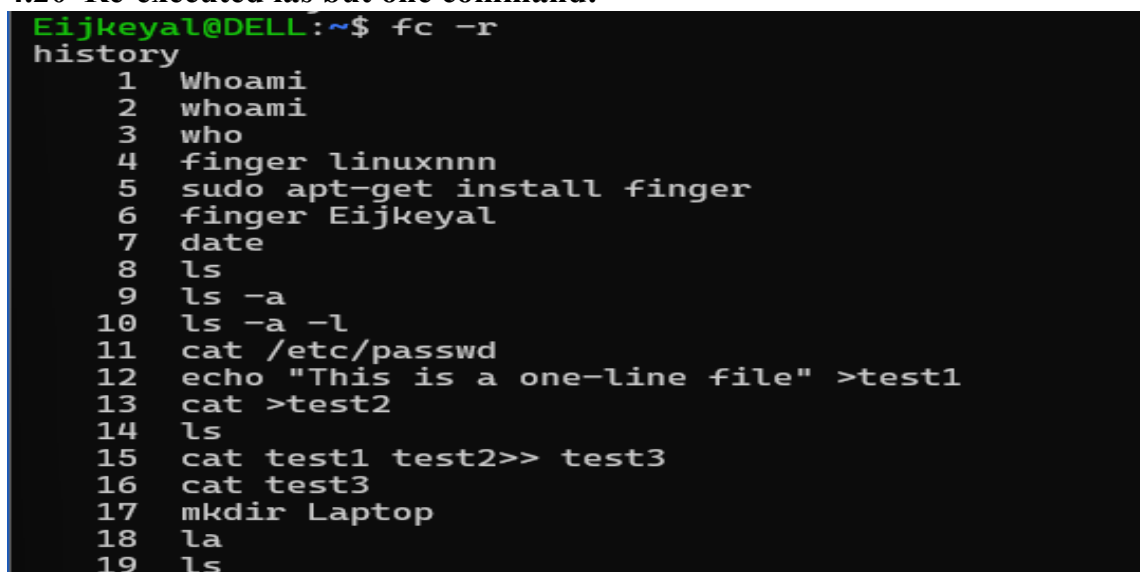
Displays all previously executed commands.



```
Eijkeyal@DELL:~$ history
 1  Whoami
 2  whoami
 3  who
 4  finger linuxnnn
 5  sudo apt-get install finger
 6  finger Eijkeyal
 7  date
 8  ls
 9  ls -a
10  ls -a -l
11  cat /etc/passwd
12  echo "This is a one-line file" >test1
13  cat >test2
14  ls
15  cat test1 test2>> test3
16  cat test3
17  mkdir Laptop
18  la
19  ls
20  pwd
21  mkdir Laptop/DELL Laptop/HP
22  touch Laptop/DELL/File1
23  touch Laptop/DELL/File2
24  rm Laptop/DELL/File1
25  ls
26  echo ""
27  echo"
```

Figure 20 Displays Previous Executed commands

4.20 Re-executed las but one command.



```
Eijkeyal@DELL:~$ fc -r
history
 1  Whoami
 2  whoami
 3  who
 4  finger linuxnnn
 5  sudo apt-get install finger
 6  finger Eijkeyal
 7  date
 8  ls
 9  ls -a
10  ls -a -l
11  cat /etc/passwd
12  echo "This is a one-line file" >test1
13  cat >test2
14  ls
15  cat test1 test2>> test3
16  cat test3
17  mkdir Laptop
18  la
19  ls
```

Figure 21 Re-execute last command

4.21 Re-execute command from three commands ago.

Executes the third previous command.

```
Eijkeyal@DELL:~$ !-3
alias nwho='getent passwd | wc -l'
```

Figure 22 Execute the third previous command

4.22 Re-execute last command starting with l

It executes the most recent command beginning with l.

```
Eijkeyal@DELL:~$ fc -e- l
lsal
total 128
drwx----- 11 Eijkeyal Eijkeyal 4096 Dec 19 06:07 .
drwxr-xr-x  3 root      root      4096 Dec  4 07:50 ..
-rw-----  1 Eijkeyal Eijkeyal 2877 Dec 18 12:46 .bash_history
-rw-r--r--  1 Eijkeyal Eijkeyal  220 Dec  4 07:50 .bash_logout
-rw-r--r--  1 Eijkeyal Eijkeyal 3594 Dec 19 06:35 .bashrc
-rw-r--r--  1 Eijkeyal Eijkeyal 3527 Dec 18 07:57 '.bashrc alias lsl='\''la -l'\''
-rw-r--r--  1 Eijkeyal Eijkeyal 3622 Dec 18 12:46 .bashrc.save
-rw-r--r--  1 Eijkeyal Eijkeyal 20480 Dec 19 06:40 c11
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 11 08:14 C11
-rw-r--r--  1 Eijkeyal Eijkeyal    0 Dec  4 08:04 File1
-rw-r--r--  1 Eijkeyal Eijkeyal    0 Dec  4 08:04 File2
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec  4 08:12 Folder
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec 11 07:33 Laptop
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 11 07:56 .local
drwxr-xr-x  2 Eijkeyal Eijkeyal 4096 Dec  4 08:03 Obj1
-rw-r--r--  1 Eijkeyal Eijkeyal  807 Dec  4 07:50 .profile
-rw-r--r--  1 Eijkeyal Eijkeyal    0 Dec  4 08:14 .sudo_as_admin_successful
-rw-r--r--  1 Eijkeyal Eijkeyal   24 Dec  5 06:00 test1
-rw-r--r--  1 Eijkeyal Eijkeyal   62 Dec  5 06:02 test2
-rw-r--r--  1 Eijkeyal Eijkeyal   86 Dec  5 06:04 test3
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 12 07:12 w7
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec 12 06:30 w7w7-1
drwxr-xr-x  4 Eijkeyal Eijkeyal 4096 Dec 12 06:30 w7w7-2
drwxr-xr-x  3 Eijkeyal Eijkeyal 4096 Dec 19 06:07 w9
-rw-r--r--  1 Eijkeyal Eijkeyal 24501 Dec 12 18:21 Workshop8
Eijkeyal@DELL:~$
```

Figure 23 Execute the last command start with l

5 Conclusion

This workshop successfully demonstrated the practical usage of essential UNIX utilities. It helps to understand file and directory management skills were enhanced using commands like `mkdir`, `cd` and `cat`. The powerful capabilities of the `grep` utility were explored for searching and analysing text data. However, command aliasing and history features were used to improve efficiency and reduce typing. To conclude this workshop improved familiarity with the UNIX command-line environment and foundation required for advanced system administration and software development tasks.